Yating Zhang

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RESEARCH INTERESTS

At the intersection of data-driven retail operations, consumer choice modeling, and global optimization, with an emphasis on developing exact solution methods for assortment planning under consumer choice behavior

ACADEMIC APPOINTMENTS

Xiamen University Xiamen, China
Assistant professor, School of Management Oct.2026 - present

EDUCATION

Shanghai Jiao Tong University Shanghai, China
Ph.D. in Management Science, Antai College of Economics and Management 2020 - 2026
Advisors: Prof. Huan Zheng, Prof. Taotao He

University of Toronto Toronto, Canada
Visiting Ph.D. Student, Rotman School of Management 2023 - 2024
Host: Prof. Ningyuan Chen

Tongji University Shanghai, China
B.Sc. in Information Systems and Management, School of Economics and Management 2016 - 2020

RESEARCH PAPERS

Assortment Optimization Under History-Dependent Utility [Link]

with Taotao He and Huan Zhen, *Operations Research* (2026).

- Existing assortment models often assume static consumer preferences, overlooking dynamic, history-dependent choices prevalent in multi-period retailing, especially in dining industry.

- Build a multi-period assortment model with cross-period history-dependent effects and cross-product substitution effects, leading to a softmax-of-exponential nonconvex structure

- Propose a lifting-based reformulation framework for the nonconvexity, improving 10.4% revenue

Assortment Optimization under Logit-Based Multi-Purchase Choice Models [Link]

with Taotao He and Zhongqi Wu

Under review at *Manufacturing & Service Operations Management*

- Consumer's multi-purchase behavior could be a mixture of intra and cross category patterns

- Propose a hypergraph representation for general multi-purchase behavior, unifying several models

- Combine linearization of the hypergraph via recursive McCormick/odd-cycle/running-intersection inequalities and convexification techniques in fractional programming to derive strong formulations

First-Choice Optimization through A Multilinear Optimization Lens

with Taotao He and Huan Zheng

Work in progress

- Use multilinear optimization as a unified approach to describe consumer choices with different preferences, obtaining strong MIP formulations for related selection problems under the random utility model

Network Revenue Management Under Regime Switching Demand

with Ningyuan Chen and Huan Zheng
Work in progress

- Leverage the regime switching information to adjust re-solving frequency of online policies

Data-Driven and People-Centric Operations: Research Opportunities in Retail Operations [Link]

with Yuanzheng Ma and Huan Zheng
Published at *Fundamental Research*, 2025

INDUSTRY PROJECTS

SmartCater: Intelligent Menu Planning (Sovel Co.) 2021-2022

Analyzed consumer choices using cafeteria transaction data, and investigated the multi-period menu planning problem considering consumer satiation behavior and operational constraints. Developed a randomized policy for industry implementation. Motivated a history-dependent choice model presented in a manuscript currently under review at *Operations Research*.

Decoding Demand: The Impact of Product Line Tiers (Jianzhijia Pharmacy) 2025

Analyze consumer choice using multi-store transaction and product line data. Estimate how product line tiers affect demand and provide insights for assortment and product line design. Motivate an order-dependent choice model in a joint optimization framework presented in a working paper.

TEACHING ASSISTANT

Modeling and Optimization (Undergraduate, AM360, SJTU) Fall 2020, 2021, 2022

- Topic: data analytics, modeling, and optimization tools for business decision-making
- Designed assignments, the final project, and a lecture on column generation and delivered a lecture on Python programming and Gurobi implementation

Optimization Methods for Economics (Undergraduate, EC200, SJTU) Spring 2021

- Topic: fundamental optimization models and tools motivated by economic applications
- Evaluated assignments and provided support for Julia coding tasks

HONORS & AWARDS

Outstanding Graduate Award, Shanghai Jiao Tong University 2026

ISCOM Best Paper Award Competition, Third Place 2024

CONFERENCE PRESENTATIONS

Strong Formulations for Logit-based Multi-Purchase Behavior

- The 16th POMS-HK International Conference (scheduled) Jan. 2026
- INFORMS Annual Meeting, Atlanta Oct. 2025

Assortment Optimization Under History-Dependent Utility

- The POMS International Conference in China Jul. 2025
- The 15th POMS-HK International Conference Jan. 2025
- INFORMS Revenue Management and Pricing (RMP) Section Conference Jul. 2024
- INFORMS Annual Meeting, Arizona Oct. 2023
- The 15th International Annual Conference of the Chinese Scholars Association for Management Science and Engineering (CSAMSE) Jul. 2023